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Q1 An online shopping platform classifies customers as High Value (H) or Low Value (L).

Customer	No. of Purchase	Time spent	Category
A	2	1	L
B	3	2	L
C	6	5	H
D	7	6	H
E	5	4	H

Solution:-

A new customer F has:

• No. of purchases = 4

• Time spent = 3

distance: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$$(F \rightarrow A) = \sqrt{(2 - 4)^2 + (1 - 3)^2}$$

$$(F \rightarrow A) = \sqrt{(2)^2 + (-2)^2}$$

$$(F \rightarrow A) = \sqrt{4 + 4}$$

$$(F \rightarrow A) = \sqrt{8}$$

$$(F \rightarrow A) = 2.82$$

$$(F \rightarrow B) = \sqrt{(3 - 4)^2 + (2 - 3)^2}$$

$$(F \rightarrow B) = \sqrt{(-1)^2 + (-1)^2}$$

$$(F \rightarrow B) = \sqrt{2}$$

$$(F \rightarrow B) = 1.41$$

$$(F \rightarrow B) = 1.41$$

$$(F \rightarrow C) = \sqrt{(6-4)^2 + (5-3)^2}$$

$$(F \rightarrow C) = \sqrt{(2)^2 + (2)^2}$$

$$(F \rightarrow C) = \sqrt{8}$$

$$(F \rightarrow C) = 2.82$$

$$(F \rightarrow D) = \sqrt{(7-4)^2 + (6-3)^2}$$

$$(F \rightarrow D) = \sqrt{(3)^2 + (3)^2}$$

$$(F \rightarrow D) = \sqrt{18}$$

$$(F \rightarrow D) = 4.24$$

$$(F \rightarrow E) = \sqrt{(5-4)^2 + (4-3)^2}$$

$$(F \rightarrow E) = \sqrt{(1)^2 + (1)^2}$$

$$(F \rightarrow E) = \sqrt{2}$$

$$(F \rightarrow E) = 1.41$$

Sort in Ascending order:-

Customer	Distance	Category
B	1.41	L
E	1.41	H
A	2.82	L
C	2.82	H
D	4.24	H

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Take $k=3$;

Using KNN Algorithm the category of customer F.

B $\rightarrow 1.41 \rightarrow L$

E $\rightarrow 1.41 \rightarrow H$

A $\rightarrow 2.82 \rightarrow L$

The customer F is classified low (L) because majority is low.

